Lecture 14: What Motivates Us: Sex (full transcript)

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Overview: This lecture reviews what evolutionary theories and recent studies in psychology can tell us about sex and gender differences. Students will hear how psychology can help explain many of the differences that exist in whom we find attractive, what we desire in a mate, and sexual orientation.

Chapter 1. Addressing Morality and Inevitability on Evolution, Sex and Gender [00:00:00]

Professor Paul Bloom: Sex is really strange. You ask people, "What's your favorite activity?" and if you ask people, particularly college students, particularly just fresh from spring break – I've seen teen movies – they'll often answer, "Sex." or some word that is synonymous with sex. But there's a kind of a puzzle about how much time we spend on sex. And it turns out there is data on this. So, people say sex is their favorite activity, but it turns out we actually know how much time the average American spends on sex. And the data I'm going to follow from was summarized in this wonderful book by James Gleick:

Americans tell pollsters their single favorite activity is sex. In terms of enjoyability, they rank sex ahead of sports, fishing, bar-hopping, hugging and kissing, talking with the family, eating, watching television, going on trips, planning trips, gardening, bathing, shopping, dressing, housework, dishwashing, laundry, visiting the dentist, and getting the car repaired. On the other hand, these same studies suggested the average time per day devoted to sex is four minutes and three seconds. [As Gleick says,] This is not much, even if the four minutes excludes time spent flirting, dancing, ogling, cruising the boulevard, toning up in gyms, toning up in beauty parlors, rehearsing pick up lines, showering, thinking about sex, reading about sex, doodling pornographically, looking at erotic magazines, renting videos, dreaming of sex, looking at fashion magazines, cleaning up after sex, coping with the consequences of sex, building towers or otherwise repressing, transferring, and sublimating .

And I like this passage because it illustrates two points, two important points. One is we don't actually spend that much time on sex. In fact, the four minutes and three seconds is an interesting number because when you do times studies on how much Americans spend filling out tax-related forms for the IRS, it's four minutes and a few seconds. But the passage also points out that regardless of the brute time we spend on it, it is extraordinarily important. Everything in life follows from it – marriage, family, children, much of aggression, much of competition, much of art and music and creative pursuits. Much of everything follows from it. If we were a creature without sex, everything would be different.

And what's interesting is, there are creatures without sex. There are creatures that reproduce by cloning. And in fact, this basic fact about people – that we fall, roughly, into males and females – is an evolutionary mystery. It's not clear why animals that are somewhat large have two sexes. From a biological Darwinian perspective, having two sexes is bizarre because each time you have an offspring you toss away half your genes. My children only have — each of them have half my DNA. If I were to clone, they would have all of it. And so, it's a puzzle how sex ever evolved.

This is not a course in evolutionary biology, and that's not the puzzle we're going to be looking at today. We're going to look at a few questions. First, we're going to talk from first a theoretical point of view and then an empirical point of view about how males and females are different. Then we're going to talk about sexual attractiveness, some research about what people find to be sexually attractive, and then we'll talk a very little bit at the end about the origins of sexual preference: why some people are straight, others gay, others bisexual, and others harder to classify.

Now, of all the topics I'm presenting, sex is one of the sort of dicey ones from an emotional point of view. These are difficult issues because sex is, by definition, an intimate part of our lives, and it matters a lot. Moreover, sex is fraught with moral implications. And since I'm talking about this from, at least at the beginning, from a Darwinian evolutionary perspective, I'm obliged to start off by dealing with some of the moral consequences and moral implications.

So, for instance, many biologists – all biologists I would say – will have argued that sexual behavior, sexual action, sexual desire is, to some extent, a biological adaptation existing to spread our genes. From that perspective then, non-procreative sex – including gay sex, sex with birth control, sex by post-menopausal women – does not serve this reproductive goal and, in some sense perhaps, is unnatural. And one might argue then, "Does this mean it's wrong?" We'll also be talking about sex differences, differences between men and women, for instance, in how much you want anonymous sexual encounters, differences between men and women in social intelligence, in aggression and empathy. And regardless of what you think about these differences, whether you think they're right or wrong or it doesn't matter, you'll ask the question, "To what extent are they mutable?" That is, if they exist through Darwinian natural selection, to what extent can we ever get rid of them? And I want to address those two issues, the issues of morality and inevitability, from the very start. And I want to start off with — for each of them have a quote by a prominent evolutionary scholar. So, the first one is by Steve Pinker in How the Mind Works. And he writes,

Nature does not dictate what we should accept or how we should live our lives. Well into my procreating years, I am so far voluntarily childless, having squandered my biological resources reading and writing, doing research, helping friends and students, and jogging in circles — ignoring the solemn imperative to spread my genes. By Darwinian standards, I am a horrible mistake, a pathetic loser, but I am happy to be that way, and if my genes don't like it they can go jump in the lake.

Pinker's point, I think, is a reasonable one. It is true that certain things we do exist to serve the dictates of natural selection, but that doesn't make them right? If you think that something is only right if it leads to the generation of more genes, if it leads to reproduction, then you're not going to think very much about birth control. You're not going to think very much about any sort of non-procreative sex. On the other hand, if you're — Moreover, if you think something's wrong if it's unnatural, you're going to think much about flying in a plane or refrigerating food or surviving a severe infection. More generally, our bodies and brains have evolved for reproductive success, but we can use these brains to choose our own destinies. Nothing moral necessarily follows from the facts of biology. That's all I'm going to say about morality. But I want you to keep it in mind when we discuss different claims about what's evolved and what hasn't.

What about inevitability? Here I want to turn to Richard Dawkins. Richard Dawkins writes,

If a child has had bad teaching in mathematics, it is accepted that a resulting deficiency can be remedied by extra-good teaching in the following year. But any suggestion that the child's deficiency might have a genetic origin is likely to be greeted with something approaching despair. If it's in the genes, it is determined and nothing can be done about it. This is pernicious nonsense on an almost astrological scale. Genetic causes and environmental causes are in principle no different from each other. Some may be harder to reverse, others may be easy. What did genes do to deserve their sinister, juggernaut-like reputation? Why are genes thought to be so much more fixed and inescapable in their effects than television, nuns or books.

I like the nuns. And the point here is what causes something is logically separate from what can reverse it. And you can think of clear cases where something is plainly genetic but is fairly easily reversed and where something is cultural and is very difficult to reverse. Here's an example. My eyesight is quite poor. The reason why my eyesight is quite poor is not due to the patriarchy, television, culture or "the man." Rather, my eyesight is quite poor due to the crappy genes Mom and Dad gave me. It is genetically determined if anything is. It is also fairly easy to fix. There are these machines where they put panes of glass in front of your eyes and help you to see better. More advanced machines known as contact lenses actually stick the thing into your eyes, and at the cost of occasional infections you come to see better. It's biologically caused but fairly easy to fix.

On the other hand, take an example of society's treatment of the obese. It turns out when we – and we'll get to this a little bit when we talk about sexual attractiveness – how thin somebody is or how fat they are; what you think of that is actually not particularly hard-wired. It varies a lot from culture to culture. But once it's in a culture, it is almost impossible to shake. So, the point, there is just that genetic does not mean inevitable, and cultural does not mean easy to fix.

Chapter 2. Basic Sex Ed [00:10:48]

Okay. That's general background. Let's start with basic Sex Ed. What's the difference between males and females? Well, don't even think penis and vagina. There are a lot of animals that have neither one. And the difference actually runs deeper. By definition, when biologists talk about this, animals that are males have a little sex cell, which carries genes and nothing else – sperm cells. Animals that are females have a big sex cell, which has genes but also food and a protective cover and all sorts of other stuff. Typically, the little sex cell is much littler than the big sex cell. This is the only erotic picture I'm going to show you today. It's a bunch of these little sperm circling around the egg. It's romantic.

But this raises a puzzle. I just described male and female roughly in terms of a size difference. Males are the smaller of the sex cells; females are the bigger. Why is it then that for so many animals males are the bigger ones, physically, and the more aggressive ones. This has been a puzzle that has occupied scientists for a long, long time. And we're pretty — there is now a pretty clear answer to it. And the answer goes like this. It is based on an idea by Robert Trivers called "parental investment." And what parental investment is, it's defined here as, any investment that's going to increase the offspring's chance of survival at the cost of the parent's ability to invest in other offspring.

So, for example, suppose an animal could create an offspring by blinking an eye and then the offspring would run off? That would be extremely little investment. Suppose another animal had to work for ten years, and during those ten years could not create another offspring. That would be a huge investment.

Trivers points out that within a species, females typically have a much higher parental investment than males. Because females have these big sex cells, they typically incubate them internally. They carry them. If they're eggs, they might have to sit on them. And hence, each potential child is a huge cost.

For males, which have the small sex cell, you don't have the same thing. For males, it might just be a few moments of copulation and that's it. If you could ask yourself, for humans, each one of you in the room, "What is the minimum effort you can do to create a child that has half your genes?" And it's apparent that the male investment, on average, is lower than the female investment. Males can choose, or might do better off in some circumstances by putting a lot of investment into their offspring, but females don't have a choice. Females, barring technological advance, have a huge investment into any offspring; not investment in the sense or hard work and effort, though there's that too. Investment in the sense that when you're — when you're pregnant with one offspring, you can't have another.

What this does is it has ramifications that percolate upwards. So, it leads to different psychologies. Males — and a single male could fertilize several females, forcing some males to go mate-less and giving rise to competition to see who can mate with the most females. For females, however, females can always find mates. So, sheer numbers don't count. But there's competition to mate with the right males, those whose offspring have the best chance of surviving. The competition now explains the puzzle we started with. It explains why males are typically larger, and often why males have evolved special weapons. These special weapons evolved for fighting other males for reproductive access. It also explains something else. Females, biologically, are choosy. And so males have to compete not merely with other males to get reproductive access but also to woo females. And so often, males have evolved special displays like this [showing a picture of a peacock's plumage], which exist only to be beautiful, only to be attractive and to attract mates.

This cold evolutionary logic was captured in this cartoon, which really does sum up a hundred of mate-selection research. The logic goes like this then: difference in the size of sex cells leads to differences in typical parental investment, leading to differences in the sorts of psychological and physiological mechanisms that evolved. Okay, that's a good story. What sort of evidence is there for it? Well, it turns out this could explain some otherwise surprising things. For instance, there should be — there are some cases where the parental investment is switched, some cases where it turns out — where the males end up with more investment than the females. And it — and the theory predicts that in these cases you should get an asymmetry.

So, in cases like pipefish, for instance, the male takes the eggs into a pouch and plugs them into his bloodstream. The females shoot off. They have less of an investment than the males. In this case, you would predict, as is true, the females should be larger, the females fight other females more than males fight males, and the females try to compete for the attention of the males. Recall the movie "March of the Penguins." We saw a clip from it, and this was in the context of discussing the emotions that have evolved toward our offspring. But remember the story and how both the male and the female have to go to tremendous lengths to protect the egg. And if one of them fails, the egg dies and neither one has it. You should then not even have to remember whether male penguins are much bigger than female penguins. You should realize they should not be, and in fact they aren't. They're about the same size because the parental is equal.

You should be able to predict the size differences and aggression differences based on differing parental investment. So for instance, elephant seals are four times — the males are enormous. They're four times bigger than the females. And this is in large part because elephant seals compete for harems of females. It's a "winner take all." Gibbons are about the same size. And this is because gibbons are pretty monogamous; they raise children together.

This illustrates something, which is, it's not always the case that male parental investment is low. There are some species, including gibbons, where it's in the male's reproductive advantage to care for the offspring. Imagine a situation, for instance, where an offspring would die if both parents didn't watch it for many years and where the effort devoted to that offspring had to be exclusive. If you focused on another family or went away, the offspring would die. In that case, you'd have equal investment. It would matter equally to the male and the female to invest in their offspring, and the cost would be the same. There's no species — it's hard to see species that have that much of an equal system, but some primates are close to it. And this raises the question then, "What about humans?" What about us? What do we know about the differences between males and females?

Chapter 3. Sex Differences among Humans [00:19:22]

Well, humans are a relatively polygamous species. Most cultures — most human cultures are polygamous. American culture is what they call "serial monogamy." So, we're not like some species of birds. We don't mate for life. We do a series of peer-bondings for some period of time. It could be for life, but indeed may not be and usually isn't. Males are bigger than females. Human males — the size estimates vary so much, but the average human male is about fifteen percent larger than the average human female. This suggests that there's some — there's been, in our evolutionary history, some male-male competition for access to females, which suggests, in turn, that the parental investment is not quite equal. Males have smaller testicles for their body size than chimpanzees, but larger testicles than gorillas and gibbons. And this suggests that there was some intermediate amount of competition for the capacity to create sperm. And this is relevant for a different sort of competition, which regards the impregnation of females that have multiple mates. And this suggests that over evolutionary history women were not wantonly promiscuous, but were not entirely monogamous either; so much so that it paid from an evolutionary point of view to evolve — males to evolve the capacity to produce more sperm than other males.

Aggression. Males are meaner. I mean I'm summarizing here. Meaner is not a technical term. Yes, females can be meaner, but males are at least more physically violent. They're more violent in the womb, in utero; they're more violent as children, and they're more violent as adults. Again, this is not to say that you can't find violent women or non-violent men. It's just on average there is this difference. They kick more; males kick more in the uterus. As children they're more involved in play fighting and violent combat-like sports. And as adults, wherever you go you will find a prison. And wherever you go you will find that prison is mostly full of men. They are far more likely to kill one another and to harm one another. Male sex hormones, like testosterone, are not the sort of thing one would want to inject in somebody unless you want them to turn kind of mean. They increase aggressiveness, both in humans and in other primates.

What about sexual choosiness? Do male humans and female humans differ in the extent to which they will favor anonymous sex? And this is relevant from an evolutionary perspective, because the parental investment theory predicts males should be more receptive to anonymous sex. Because for males, to impregnate somebody else might fortuitously lead to another offspring; it might be good for you and doesn't carry the sort of harm that females, on the other hand, have to be very picky. Because they have to choose carefully. Remember, these systems evolved before birth control and vasectomies and so on. So, what do we know cross-culturally and psychologically?

Well, prostitution is a universally, or near universally, male interest. There are male prostitutes, of course, but contrary to some various fantasies and sitcoms, they cater to male customers. Pornography is a human universal. In every society, males have done some sort of depictions of naked females for the purposes of arousal. Often they carve them into trees or do sort of sculptures. One of the weirdest findings in the last decade or so is that this extends as well to monkey porn. And so, some scientists at Duke set up a situation where monkeys could pay in fruit juice, by giving up fruit juice, to look a picture either of the female's hindquarters or of a celebrity monkey, a socially dominant monkey, some sort of combination of People Magazine and Penthouse. And so, there's some interest in this even by non-human primates.

What about actually preference for sexual variety? Well, you can get at this in different ways. There is what biologists describe as the "Coolidge Effect." I have this here [on a slide]. And the Coolidge Effect is based on President Calvin Coolidge. And it's a story about Calvin Coolidge and his wife, who were being shown around a farm separately. And the person showing around his wife pointed out that there were a lot of hens; she noticed that there were a lot of hens but only one rooster. And she asked the guy showing her around, "Is one rooster enough?" And the guys said, "Well, you know, the rooster works very hard. The rooster has sex dozens of times a day." And she said, "Well, be sure to tell that to the president." The story goes, the president went around, the guy tells the story to the president. The president asks the man, "Huh. Has sex dozens of times each day. Same hen every time?" The guy says, "No, different hen every time." And he says, "Tell that to Mrs. Coolidge."

Now, there are two responses to this sort of story, and they're both kind of negative. One thing is, "Well, everybody knows males prefer anonymous sex with strange women. Duh." The other response is, "That's sexist claptrap." You might think — you might be a male and say, "That's not me." You might know males and say, "The males I know are not like that." So, how do you find out? Well, there are indirect measures, such who goes to prostitutes. But there are also fairly direct measures. One fairly direct measure is you could ask people in anonymous surveys. So, in fact, I'll give you some anonymous surveys. I'm not going to ask people. And you just ask them. So, for instance, I want everybody to consider this question. How many sexual partners do you want to have in the next month? What is it — we're coming up to April. How many sexual partners do you want in April? Next two years? Take many of you through graduation. When you leave Yale, what do you want — like, "I had X sexual partners, and that's what I wanted." Or your lifetime? We get people to answer these questions.

Professor Chun last year in this course had clickers, and he got people to do it. We are not so high tech, so we'll just do it in our heads. But here is the way the answers come out. Women say less than one in the next month. That doesn't mean they want less than one; that means many of them — many of them say zero, some say one and so on. One — four to five. Men — two, eight, eighteen. You can ask other

questions from this population. So, you could ask, "Would you have sex with a desirable partner you have known — so somebody really desirable — for a year; women say yes, six months — unsure, week or less — no. Men [slide reads "yes, yes, yes"] [laughter] — and with men you could get a majority going to five minutes.

This is all Q & A, pen and pencil sort of things. Some brave scientists have actually done experiments. And in one experiment somebody — I don't, you know, this is the sort of thing which you probably wouldn't do nowadays. This work has been done ten years ago, where they have an incredibly attractive man and an incredibly attractive woman and they approach people on campus. They're not from campus; they're actors brought in. And they go to people, to strangers, and they say, "I've been noticing you around campus. I find you very attractive. Would you go out with me tonight? Would you come over to my apartment tonight? Would you go to bed with me tonight?" The experiment you wouldn't think anybody would've done has been done, and women about — a very attractive man, over half of the women approached say, "Yeah, I will. [go out with you tonight]" Very few agree to this ["Would you come to my apartment tonight?"], and nobody agrees to this ["Would you go to bed with me tonight?"]. For men, the data are like this, they go up to there and then up to there [they say yes 50%, 69% and 75% of the time respectively]. In this study, the twenty five percent of males who said "no" apologized profusely, and they said, "Oh, you know, my fiancé's in town, and [unintelligible]."

What about behavior? Well, you — if we're interested in sex differences, you can't actually figure out what people want, male female differences, by looking at simply at the average number of times people have sex because if males and females have different priorities, then heterosexual sex is a compromise between two groups of people with competing interests. What's a more clear reflection then is gay sex between two women or between two men, because then you get a pure reflection of sexual desire. Now, the data here tend to be very messy. Again, they're survey studies but by and large every study done tends to find a difference in the expected direction, which is that females tend to be — lesbians tend to be much more monogamous than gay men. Some studies prior to AIDS – this was many years ago – found gay men to be extremely promiscuous, often having over a hundred or over a thousand partners. You wouldn't find this sort of promiscuity in females. And a way to think about this is, what these gay men are doing is exactly what your average heterosexual man would do if he had that degree of willing females who were as willing as he was. And this all suggests that there's some sort of difference along lines expected in sexual choosiness in humans.

What about sexual attractiveness? What about mate preference? What do we find attractive? Well, unlike the choosiness studies, here we actually have some pretty good cross-cultural data. So one study, for instance, was done in 10,000 people from thirty-seven countries, asking people, "Who do you want to be with?" And there are different studies, some of them asking, "Who do you want to marry?" Other studies, "Who do you want as a mate? Who do you want as a sexual partner?" And one main finding is kind of reassuring, everybody likes kindness and intelligence, or at least everybody says they like kindness and intelligence. These are valued pretty highly.

But at the same time, there are sex differences. Females focus more on power and status and more on interest in investing in children. And think about that from an evolutionary point of view and it makes sense. It doesn't matter hugely, from the standpoint of reproduction, how old the man is. The difference between fifteen and twenty-five and thirty-five and forty-five may matter a lot for his status

in the community, his physical strength, his lifespan but from the standpoint of his sperm it doesn't matter hugely. Later on there's a drop off and it does begin to matter, but it doesn't matter hugely. What does matter is his interest in being a good father, in protecting you from predation, from murder, from assault by other people, and in taking care of the kid. Women's brains are wired up to find males with those properties.

Similarly, males focus a little bit differently. They're more interested in all of these things, but also on the ability to have children. So, from an evolutionary point of view, there's actually a very big difference between a twenty-year old and a fifty-year old, from a male standpoint looking at a female, because the one can have offspring and the other cannot. So, this is a difference.

Chapter 4. Beauty: An Average Face [00:32:16]

But what I want to focus more on right now is back to another similarity. Everybody likes beauty. And I want to devote a little bit of this lecture to talking about physical beauty. Physical beauty, as these beautiful people say, is a curse. So she — she's like a big model, a supermodel, maybe even a super-supermodel — points out the arbitrariness of finding her devastatingly beautiful. Famous actor points out how frustrating it is that people only ignore his accomplishments and focus merely on his physical beauty. This is very frustrating. So what is beauty? What does this mean we say we find — you know, yeah, they really are very attractive people. What is it about that that makes you look and say, "Yeah, that makes sense?" Well, we kind of know the answer. We know some universals.

Beauty seems to signal two things. Beauty seems to signal youth — I mean, not pre-school youth, but youth like sexually mature but young. And so the cues we find beautiful are cues to that – large eyes, full lips, smooth, tight skin. Beauty signals something else. Beauty is a marker for health. And so what we find beautiful, things like the absence of deformities, clear eyes, unblemished skin, intact teeth – that's very big – and an average face. And that last part might seem a little bit strange. What would be so good about an average face? And there are different answers to that, but one answer is, an average face, on average, should be considered attractive because any sort of deformities are variations from the average. And if you average every face together, you get a face that — where nothing bad has happened to it. There's no distortion, there's no deviation. As one gets older, the face gets less symmetrical and so on. Average-ness seems to factor out all the bad things that could happen.

Good theory. How do we know it's true? Well, there's a photo roster that comes — that I have access to for this class. So, I can look at each of your pictures, and I will make you a bet about who has the most beautiful face in this course. The bet is it would be all of you. Aw. Wouldn't it be funny if I shouted out somebody's name? And you know, A) I don't have the energy to do this, and B) it would probably violate four hundred different privacy laws or whatever. But if I took all those faces and morphed them together, I would get a very pretty face. And how do we know this? Well, people have done this. They've done it with — so look at the faces from here to here. And if you are like most people, you see as you're going to the right they're looking better and better and better. It's subtle, but it's actually not so subtle that babies don't notice it. The same researchers who constructed this — these face — these average Caucasian faces, male and female, have shown these faces to babies and find that babies that prefer to look at average faces — suggesting that our preference for averaging is not the product of culture but rather is to some extent hard-wired.

These two people don't exist. They're computer composites. They're a heavily averaged male face and a heavily averaged female face, both from a Caucasian data sample. They don't look bad right? They're good faces. They don't cheat. So the hair, for instance, is identical — so they don't — you can't use hair cues. But they're pretty attractive. But the story of attractiveness does not end there. How do you get a better than average face? What can you do to these faces, these average faces, and make them look even better? Well, I'll have a vote. Who's prettier? Who says the one on the right? Who says the one on the left? Left is average face, and there might be variation in this class. There are definitely variations in what people favor. This is a feminized version of the average face where certain prototype features were made more feminine than average to cue this as more of a sexual object.

This is more complicated. Who thinks face A is more attractive? [face A has exaggerated chiseled jaw and square chin] Who thinks face B is more attractive? Okay. Most people like face B. The exception is, and this has been statistically replicated, I think, now in three labs. Face A is preferred by women who are ovulating, and the story about why is complicated and will take us beyond the scope of this class. But currently the idea is that this [face B] is a really handsome guy; he's young, he's healthy, he looks strong, good provider; this guy [face A] is really hot, and he may not be a good provider and everything, but I'm sure he has wonderful genes. So, the idea is that one should have sex with him [face A] and then have him [face B] raise the kids.

Chapter 5. Social Factors for Sex Differences [00:38:24]

We've talked so far about things, about sex and sexual attractiveness largely from a biological perspective, looking at universals. And in fact, there are some universals in what men and women have in common and what distinguishes men and women. And in some of the sex differences, particularly related to aggression and mate preference, seem to be universal. They seem to show up to some extent across every culture you look at and, hence, are likely candidates for biological adaptations. But there are other sex differences that people are aware of where the origins are far less clear. And I think that intelligent, reasonable people could disagree about this, but I am personally quite skeptical about the extent to which these reflect biology. I'll mention them to capture the debate, but the thing to keep in mind here is that biology, natural selection, is one reason why the men in this class might differ from the women in this class. But of course, there are other, social, factors.

Babies are treated differently. There have been many studies where you take a baby and swaddle it in blue and describe it as a boy versus swaddle it in pink and describe it as a girl, and people treat it differently when they think it's a boy than when they think it's a girl. You're treated differently too. It matters a lot — and there's study after study suggesting, for instance, that when you send an email or a job application or a paper to a scientific journal, it matters whether it has the name John Smith on it versus the name Joan Smith. It matters because people have different expectations and different reactions to males versus females. Some if you may have firsthand experience with this if you're a man with a name that could be taken as a woman's name – friend of mine is named Lynn, and often people think he's female – or if you're a woman who has a name that could be taken as a man's name, or if you have a name sufficiently foreign to Western ears that people can't easily tell. You'll often find people saying, "Oh," people are high-fiving each other there [referring to students in the class] — you'll often find some degree of surprise and some degree of people saying, "Oh my, I didn't know you were a man.

Now I will treat you differently." And so, these social factors could play a role in explaining some male and female differences.

Also, there are the facts of gender self-segregation. So here something very interesting happens developmentally. Males segregate with other males; females segregate with other females — for a period lasting, it depends on the culture, but say from age four to age eleven. This self-segregation might exaggerate and enhance sex differences. It might be, for instance, as Eleanor Maccoby has proposed, that boys are slightly more aggressive than girls. But then boys go into groups of boys, and that enhances and exaggerates their aggression while the girls' non-aggressive behavior is enhanced and exaggerated in different ways by them falling into girls' groups.

So, what sort of differences are we talking about when we say we're not sure of their cause? Well, one difference is one in empathy. So Simon Baron-Cohen wrote a wonderful book called The Essential Difference where he argues that men are by nature less empathetic, women are by nature more empathetic, and that this is a core sex difference. So, what do you know? Well, what's the source for this? One thing is men are more violent. Simon Baron-Cohen describes violence as the ultimate act — murder as the ultimate act of a lack of empathy. There's some relationship between how much testosterone you have in your system and how social you are – more testosterone, less social. Boys tend to be less empathetic than girls, and there's some evidence, though it's not conclusive, that boys do worse than girls on social cognition theory of mind tasks. That's what I have here, though that is quite debated.

And the biggest effect, which isn't debated at all, is problems with empathy, problems of social cognition, are much more frequent in men than in women. So, these disorders like autism, Asperger's Syndrome, conduct disorder and psychopathy are predominately male. And Simon Baron-Cohen suggests that — basically, he has this slogan where he says, "To be a man is to suffer from a particularly mild form of autism." That males are just socially clueless relative to women. Final bit of trivia — This is Simon Baron Cohen, who is a very famous developmental psychologist, but his cousin Sasha Baron Cohen is far more famous. Another debate is a debate concerning sex differences in the capacity for math and science.

A few years ago there used to be a president of Harvard known as Larry Summers. There are so many reasons to hiss at this point. And Larry Summers is no longer president of Harvard for various reasons, but one reason was this quote, which included in his speculations about sex differences, "...in the special case of science and engineering, there are issues of intrinsic aptitude, and particularly of the variability of aptitude..." He argued, or suggested, that the under-representation of women in the sciences in academia is because of an intrinsic aptitude difference; women are, on average, less biologically predisposed to do this sort of reasoning. The variability point is that he wasn't suggesting that there's just a difference on average. In fact, he agreed that the average skills of men and women are identical. The claim is that males show more variation. This means that there are more male retarded people and more males who are just horribly bad at this, but it also means there are more male geniuses. And he suggested that this plays a role.

This, as you can imagine, proved to be an extremely controversial claim, and rather than go through it – because it would take me a class to treat the pros and cons of this argument responsibly – I'm going to refer you to a wonderful debate between Steve Pinker, who was quoted earlier, and Liz Spelke who's

one of the big infant cognition people. And we spoke a lot about her work earlier on in the course. And they have a wonderful debate between two of the smartest people I know on The Edge, which was done at Harvard about a year ago and is on video here [http://www.edge.org/3rd_culture/debate05/debate05_index.html]. So, if you're interested in sex differences and different theories about the mechanism of sex differences, this is where you should go.

Chapter 6. Sexual Orientation [00:45:53]

Finally, and a final topic, some of us, about 98% — and the numbers are very difficult to pin down. Maybe it isn't 98%; maybe it's ninety-seven, maybe it's ninety-nine. Let's say 98% of women are sexually attracted to men. About 96% of men are sexually attracted to women. And the numbers vary and it's very difficult to estimate it properly. As you could imagine, there are all sorts of problems with this sort of research. But there's some proportion of the population that's exclusively homosexual — some proportion of the population of men who are only attracted to other men, some proportion of the population of women who are only attracted to other women. When people talk about sexual orientation here, it's important to realize we are not talking here about behavior. There are all sorts of reasons why somebody might have sex with somebody of the same sex. You know, they might be bored. You may, you know, be experimenting, be whatever. The question is, "What do you want to do?" All things being equal, what sort of person — if you could be sexually or romantically involved with any person, who would it be? And most people are heterosexual. There's a considerable amount that varies cross-culturally of people who are bisexual. But the real puzzle is exclusive homosexuality. So, why?

Well, nobody knows. We know some reasons, some answers are probably not right. It is not the case, almost certainly – maybe there are some exceptions – but it is not the case that people choose their sexual orientation. I'm not going to do this in this room, but if you asked people to raise their hands as to how many people decided who to become sexually attracted to, very few people would. Part of the issue rises in the fact that people who are gay are often extremely discriminated against, and they have no wish to be gay. They might even think it's morally wrong for them to be that way. That makes it implausible that their sexual orientation is a conscious choice.

What about experience after puberty? So, there is a view that keeps coming up over and over again in the literature that people who are gay have in some sense been seduced by people, by other people — or something happened to them afterwards. This seems unlikely. There are in particular the seeds of sexual orientation later on in life seem to show up quite early in life. Again, the studies are sort of suspect, but there's some reason to believe that people who are gay and people who are straight are different long before they hit puberty with regard to their sexual and romantic fantasies.

You would now expect me to say, "Well, being gay and being straight is built in. It's hard-wired. None of these stories seem right. It seems to be built in." And the answer to that is, sort of. So, if you do the standard behavioral genetic tests, and you by now know how to do them — you'd look for differences between monozygotic and dizygotic twins, you'd do the adoption comparison — you know adopted siblings and biological siblings. The answer is yes, you find that there is some sort of genetic predisposition towards homosexuality. But it can't be entirely genetic. One reason why it can't be entirely genetic is, if I'm gay and I have an identical twin, the odds that my identical twin will be gay — it's about fifty percent. Those are very high odds compared to the average in the population. But if it

was truly genetic, entirely genetic, what should the number be? A hundred percent – he's my clone. He should be exactly as I am. And it's not. So, we know then that some sort of experience, possibly prenatal experience, is what explains it.

Why is it so – I said before this is a huge puzzle – why is it such a huge puzzle? Well, exclusive homosexuality is an evolutionary mystery. Again, do not think that this carries any moral weight to it. What it does mean though is that it doesn't seem to follow as a biological adaptation. The puzzle is not why is it that some men have sex with men. That's not a big puzzle. Maybe they have sex with men as some sort of recreational things or pair bonding or whatever. That's not the puzzle.

The puzzle is why are there some men who don't want to have sex with women? Similarly, why are there some women who don't want to have sex with men? From an evolutionary adaptive standpoint, you would think that the genes that give rise to such a behavior would be weeded out because creatures with that behavior typically, putting aside modern technology, don't have offspring. And that's what makes it such a puzzle. So, your reading response for this week is "solve that puzzle." I know I said early on in the course that reading responses would be really easy and just require you reciting back things, but that proved to be too boring. So, just solve this deepest of all puzzle. The thing in brackets at the end is very important. Your account, whatever it is, should bear some relationship to the facts as discussed in lectures and readings. We have about five more minutes.

Chapter 7. Question and Answer on Sex and Gender [00:51:39]

Any questions or thoughts? Yes?

Student: I like your leather jacket.

Professor Paul Bloom: Thank you very much. She likes my leather jacket. Any questions or thoughts, just like that one? No. Yes?

Student: My question's not exactly like that one, but in other animals do they — is there similar data on other species?

Professor Paul Bloom: On sexual preferences? That's a very good question because certainly your answer to the origin – give me two more minutes – certainly your answer about the origin of sexual preference in humans will be informed by the question of cross-species data. What we do know is that there are many animals that engage in homosexual behavior; they engage in sex with members of their own sex. What I don't know is whether you get exclusive homosexual behavior. So, I don't know what the rate is in nonhuman primates, for instance, of primates who do not want to have sex with members of the opposite sex. Okay, I'll see you all Wednesday.